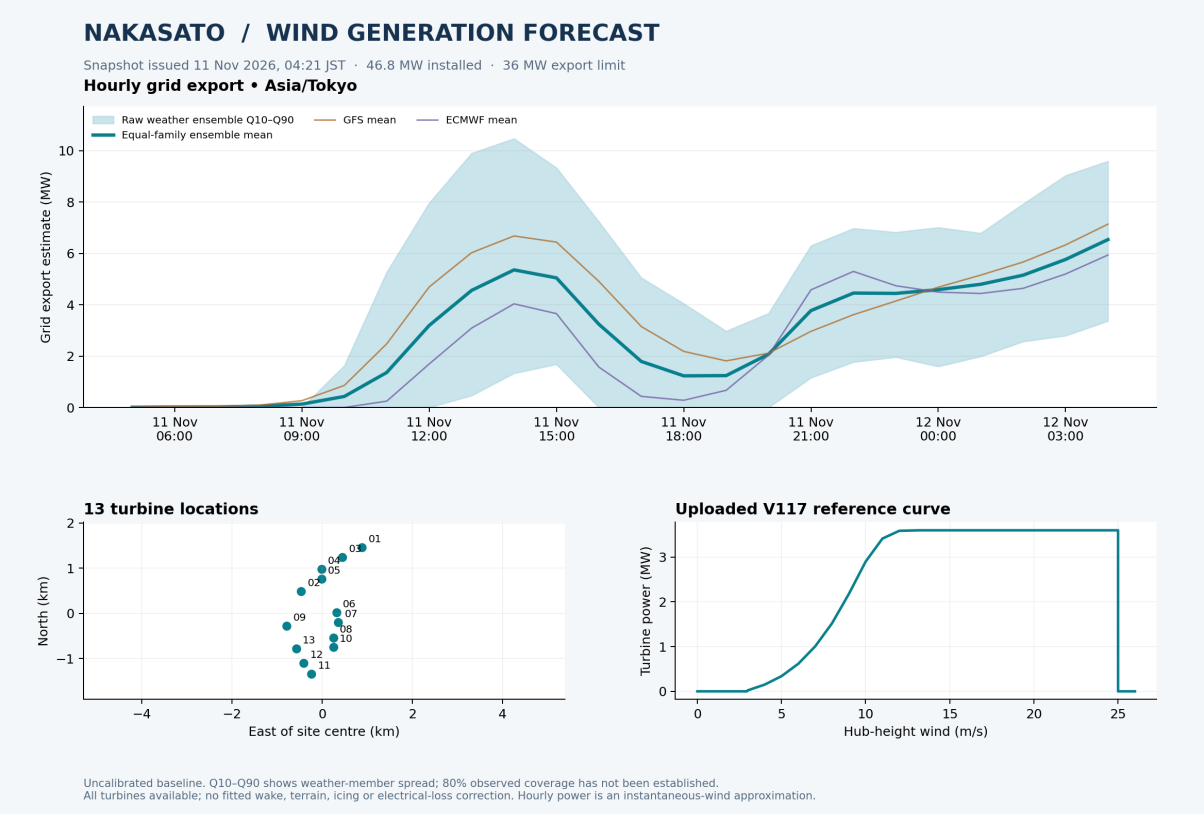
NAKASATO
Wind generation forecasting

A runnable ensemble model for your 13 turbine locations

Turbines	Installed capacity	Export limit	Weather members
13	46.8 MW	36 MW	82 across 2 families

Built now: hourly wind-to-power forecasts, joint farm scenarios, export limiting, raw Q10/Q50/Q90, an Excel workbook and a reproducible Python notebook.

No SCADA required: wind inputs come from the weather API and generation comes from your power curve. Historical data is only needed for later site calibration and measured accuracy assessment.



Included weather snapshot: 11 Nov 2026, 04:21 JST. It covers 24 hourly timestamps, from 11 Nov 05:00 to 12 Nov 04:00 JST. Run the live command to refresh it.

How the forecasting model works

Stage	Implementation
Weather inputs	GFS and ECMWF ensemble speed/direction at 100 m for every map coordinate; JMA MSM at 10 m as a separate comparison.
Hub-height adjustment	v_hub = v_reference x (116.5 / reference_height)^0.14. The exponent is configurable and has not been fitted to this site.
Power conversion	Convert every member through the sorted uploaded 3.6 MW curve. Apply explicit cut-in, cut-out and an optional hourly restart approximation.
Operations	Apply known turbine availability and power limits. Default: all available, local wind multipliers=1, electrical losses=0.
Farm export	Preserve family/member across turbines, sum turbine power, apply losses, then cap each scenario at 36 MW.
Ensemble output	50% weight per weather family; equal member weights within a family. Calculate mean and weighted CDF quantiles after power conversion and export limiting.

Why average scenario power?

The turbine curve is nonlinear. Converting average wind generally gives a different answer from averaging converted member power. The export cap adds another nonlinearity, so it is also applied before averaging. A family with more members does not receive more total weight.

What the input audit established

Your workbook has 522 rows, covering 0-26 in 0.1 steps at two density values. Both density curves have identical power values. This baseline uses 1.225 and no further density correction. Units are inferred as m/s, kW and kg/m3 and should be confirmed.

The curve reaches 3,600 kW, gives rated output at 25.0 m/s and zero at 25.1 m/s. The model never interpolates across that shutdown jump. After a high-wind stop, the optional hourly state model waits until wind is at most 23 m/s; initial state is assumed running.

Shared weather does not prove shared local conditions

All 13 locations returned identical wind trajectories within each source in this snapshot. The map spans approximately 3.01 km. Wakes, terrain and operating differences can still matter. Initial turbine predictions match until observations support local corrections.

Q10-Q90 represents raw weather-member spread, not demonstrated 80% observed coverage. Hourly generation and energy values approximate hourly mean conditions from instantaneous wind samples.

Optional future calibration and evaluation

The current 24-hour ensemble runs without SCADA. The optional second stage is implemented but has not been trained on your plant. It combines the physical mean with Random Forest, histogram gradient boosting and Ridge residual corrections. Nonnegative weights sum to one and are learned from chronological out-of-fold predictions.

Step	Validation rule
Development	Use increasing time windows. Before each validation block, remove training targets and labels unavailable at that block's first issue time.
Weight learning	Fit weights only to development out-of-fold predictions. Keep the final 20% of issue times untouched.
Held-out reporting	Report MAE, RMSE, capacity-normalized errors and bias, with a lead-time breakdown and origin-time persistence comparison.
Production refit	After recording holdout results, fit a separate model on available labels. Keep the evaluated model so its reported result remains reproducible.

Required generation data

Column	Meaning
valid_time_utc	Beginning of the target hour, in UTC.
turbine_id	Map/SCADA ID from 01 to 13.
power_kw	Actual mean active power over that hour, in kW.
observed_at_utc	When the hourly label actually became available to the forecasting system.

Aim for 6-12 months, preferably a full year, paired with forecasts available at each historical origin. Convert subhourly SCADA to correctly aligned hourly means; distinguish incomplete data from zero production. Farm export validation also requires the export meter.

What was tested

14 software tests passed. They cover interpolation, shutdown/restart, family weights, scenario aggregation, grid limiting, future-information guards, persistence, snapshot consistency, and a complete synthetic train/save/predict workflow. The synthetic fixture verifies software only and supplies no plant accuracy evidence.

How to answer the professor's ensemble question

Evaluate persistence, individual weather/physical forecasts, each residual learner and the stack on identical held-out periods. A later LSTM comparison should use those same inputs and splits. Report results by horizon and season. Ensembles are a candidate to test; superior performance is not guaranteed.

Run it and continue the project

Open the complete project

Extract Nakasato_Wind_Forecast.zip. Open Nakasato_Wind_Forecast.ipynb for the guided workflow, or use the terminal commands below from the extracted project folder. Python 3.12 was tested.

```
python -m pip install -r requirements.txt

python -m wind_forecast forecast --replay \
  examples/weather_snapshot --out runs/replayed

python -m wind_forecast forecast --hours 24

python -m unittest discover -s tests -v
```

Files to use

File	Purpose
examples/forecast/Nakasato_forecast.xlsx	Hourly farm/turbine forecasts, JMA comparison, source grid audit, curve and scenario table.
examples/forecast/run_metadata.json	Run status, assumptions, source requests, hashes, member counts and timestamp definition.
MODEL_CARD.md	Full formulas, model definition and interpretation.
DATA_REQUIREMENTS.md	SCADA schema, aggregation/time rules and validation design.
data/turbines.csv + config.json	Turbine settings, wind corrections, export cap and other configuration.

Primary sources

Open-Meteo Ensemble API

Open-Meteo JMA API

Open-Meteo Single Runs API

Vena: installed and grid capacity

Vena: hub height

Vestas V117-3.45 MW specification

Research context: ensemble-to-power forecasting

The project preserves your original power-curve workbook and the retrieved site KML. Confirm current turbine configuration, source units and meter definitions with the operator before interpreting forecasts as operational commitments.